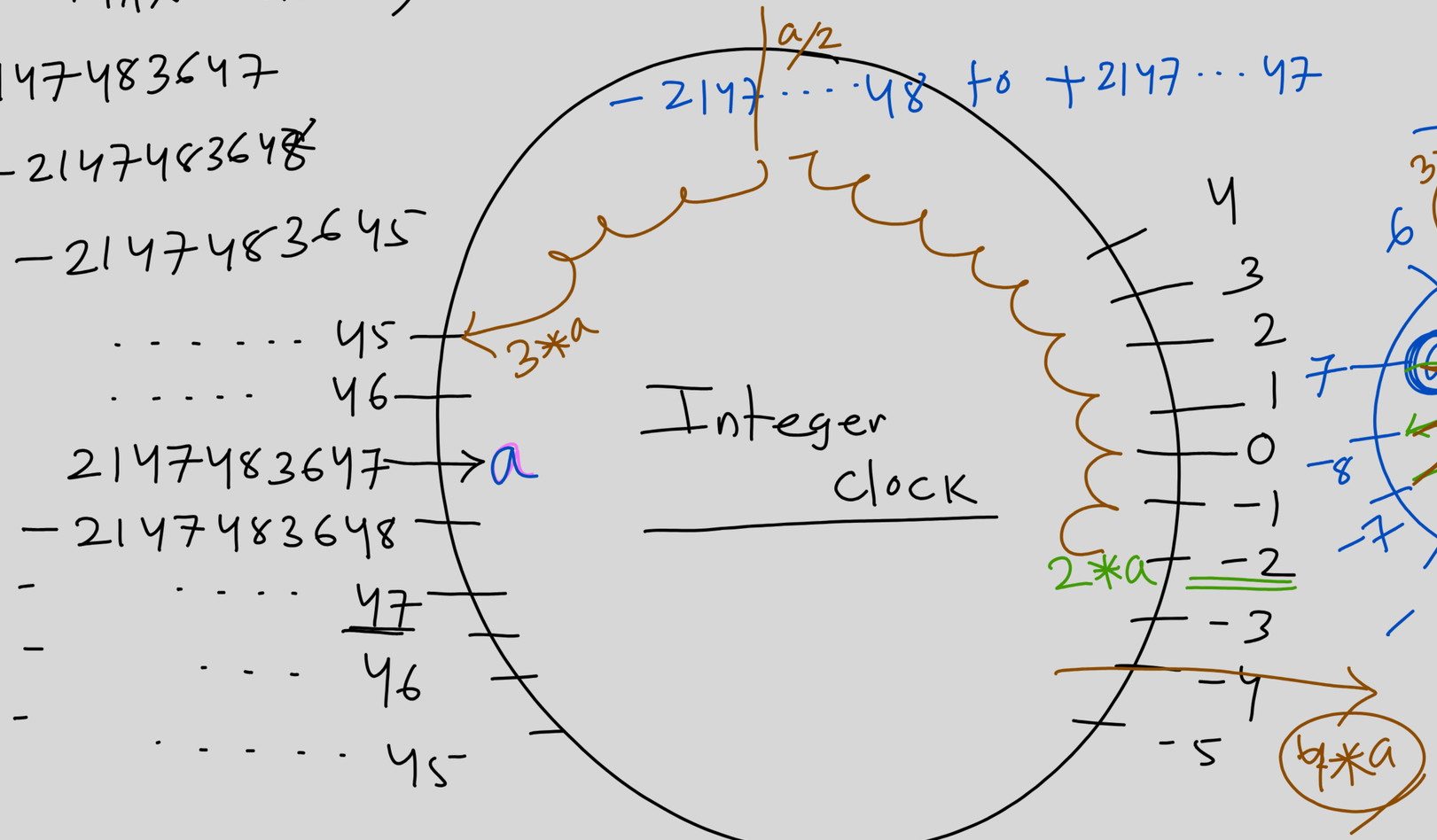
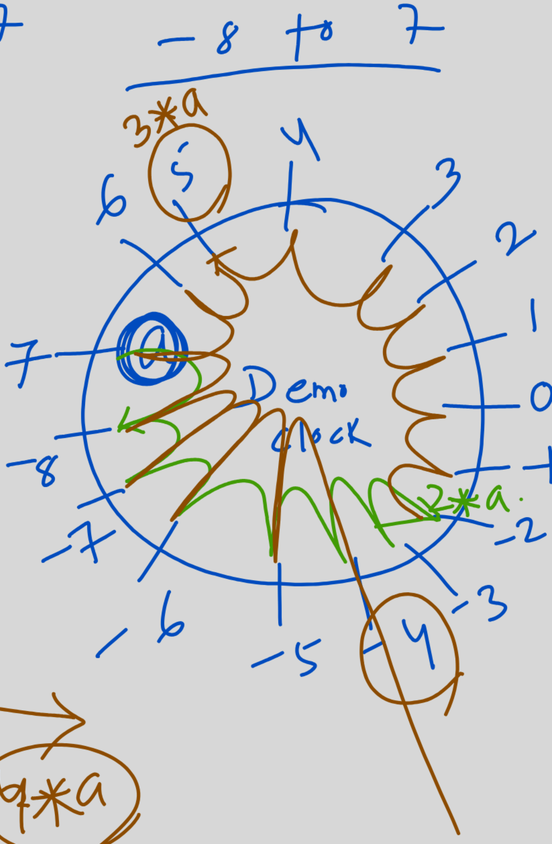
```
int a = Integer.MAX_VALUE;
s.o.pln(a); → 2147483647
s.o.pln(a+1); → -2147483648
s.o.pln(2-a); → -2147483645
s.o.pln(-2-a) → 2147483647
s.o.pln(2*a) → -2
```

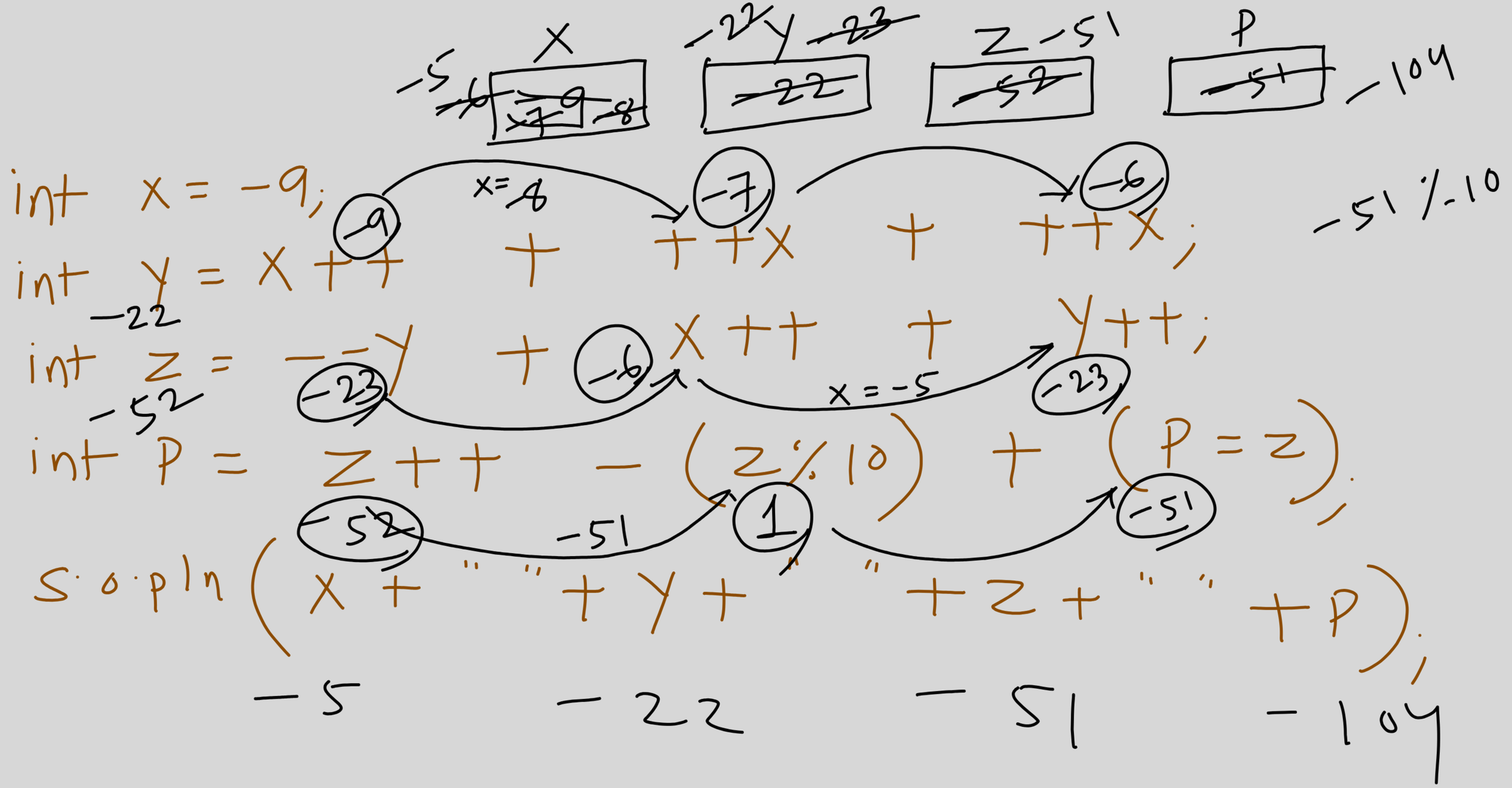




ICP

Quiz

Q2


X
Y
Z
P
-5
-9
-8
-22
-23
-52
-51
-104
int x = -9;
int y = x++ + ++x + ++x;
int z = --y + x++ + y++;
int p = z++ - (z%10) + (p=z);
System.out.println(x + " " + y + " " + z + " " + p);
x=-8
x=-5
-51%10
-5
-22
-51
-104

Looping / Iteration. :- Repeat a stmt/set of stmts.

① → while

② → do-while

③ → for.

*Keywords :- do, while, for.

Steps in a Loop

① initializtn

② Condn (where to stop)

③ Incr/Dcr

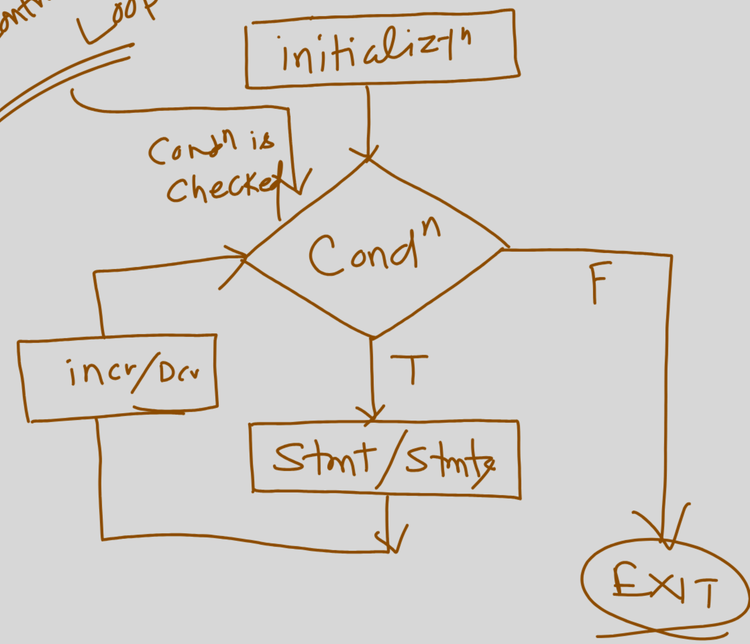
Syntax

```
initializtn
while (Condn)
{
    stmt/stmts that you want to Repeat
    incr/Dcr
}
```

Scope of while.

Body of the Loop.

entry controlled Loop.



Q1

Input: 'n'

Output: 1 2 3 . . . . n

① initializ^{n} :- i = 1

② Cond^{n} :- i <= n

③ incr/Dcr :- i++

```
int n = sc.nextInt();
int Prod=1;   int i=2;
while (i<=n)
{
  Prod = Prod*i;
  i = i+2;
}
s.o.pln (Prod);
```

initializ^{n} → int Prod=1; int i=2;

while (i<=n) ← Cond^{n}

i = i+2; ← Java starts.

Execut^{n} pattern / Dry Run

n = 6

i = 1 → i<=6 → Print i → i++

1	→ 1<=6	I→	1 → 2
2	→ 2<=6	I→	2 → 3
3	→ 3<=6	I→	3 → 4
4	→ 4<=6	I→	4 → 5
5	→ 5<=6	I→	5 → 6
6	→ 6<=6	I→	6 → 7
7	→ 7<=6	F	EXIT

Let

n = 6

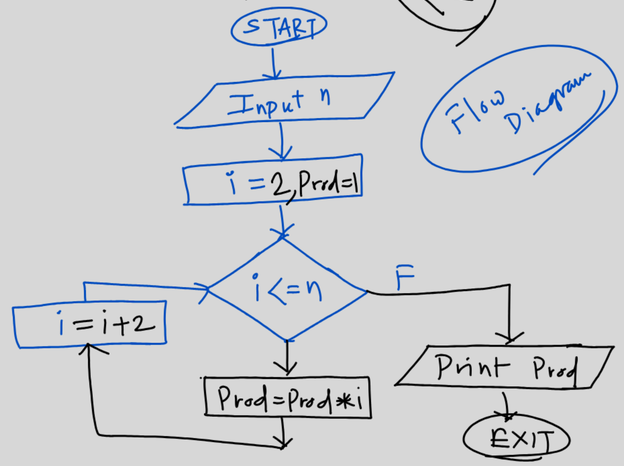
1 2 3 4 5 6

i

i = 7 → stop.

i (>) n → stop

Complement (<=) → Run.



n = 10

i = 1 → 3 → 5 → 7 → 9 . . .

+2 +2 +2 +2

Let n=10 Print Product of all even and odd Separately.

Range 1 2 3 4 5 6 7 8 9 10

Prode	Prodo
1*1	1*2
1*3	2*4
3*5	8*6

```
int n=sc.nextInt();
int i=1, sum=0; boolean P=false;
while(i<=n)
{
  if(i%2==0 || i%7==0)
  {
    P=(i%2==0 && i%7==0)
    if(!P)
    {
      sum=sum+i;
    }
  }
  i++;
}
```

